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Systematic Review

Generative AI for Learning Individualization and Student Support in Sustainable Higher Education: A Systematic Review of Creative Agency, Equitable Learning, and Implications for SDG 4 and SDG 10

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Abstract

Generative AI (GenAI) promises individualized support in higher education, yet sustained learning and equity benefits remain uncertain. This systematic review examined learning individualization, student support, creative agency, equitable learning, and implications for Sustainable Development Goals (SDGs) 4 and 10. Prospectively registered (INPLASY202660012) and reported following PRISMA 2020, it searched Scopus, ERIC, ScienceDirect, and the MDPI Journals Platform. Sixty-one studies represented by 63 reports underwent independent dual appraisal using the Mixed Methods Appraisal Tool (criterion-level Cohen's $\kappa = 0.664$) and narrative synthesis. Learner-directed individualization appeared in 46 studies; five documented system-driven adaptation. Creative-agency findings were mixed in 41 of 60 studies, combining ideation and autonomy with overreliance concerns. Equitable-learning findings were mixed in 17 of 35 studies, with accessibility and linguistic support alongside unequal access and benefit. Evidence frequently relied on self-reports, with limited long-term follow-up. Seven studies explicitly framed findings around SDG 4, one around SDG 10, and none measured official SDG indicators. Core themes remained represented after 39 studies were temporarily omitted in sensitivity analysis, without establishing robustness. GenAI offers flexible support, but sustained learning gains and reduced educational inequality remain unestablished. Future research should assess learning, compare groups longitudinally, and specify SDG targets.

Keywords: generative artificial intelligence; higher education; learning individualization; student support; creative agency; educational equity

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1. Introduction

GenAI has opened new possibilities for learning individualization and student support in higher education [1]. Students from many countries have reported using ChatGPT to generate ideas, summarize texts, and locate research articles [2]. At several universities, students also expressed expectations of personalized learning support and writing assistance, although these perceptions do not establish improvements in learning outcomes [3]. These possibilities coexist with concerns about the accuracy of generated content, privacy, and assessment practices [4]. Academic integrity presents an additional challenge: generated text may be submitted as students' own work without being flagged by conventional plagiarism detection systems [5]. An analysis of university policies described the guidance they proposed on ethical use, authentic assessment design, and GenAI literacy training; equity, however, received less attention, and the authors noted gaps in data protection and equitable access to these tools [6].

Beyond these uses and institutional responses, reported applications of GenAI include personalized guidance and self-directed study, alongside concerns that heavy reliance on such support could limit students' ability to develop ideas and collaborate [7]. In higher education, the adaptation of these tools to educational contexts and the evaluation of teaching strategies remain priorities for further investigation [8]. Creativity, autonomy, and critical thinking have also received attention, but their interconnections need closer study, while automated assistance may encourage learners to delegate cognitive effort [9]. Sustainability calls for comparable scrutiny: the contributions proposed for GenAI to sustainable higher education still require empirical testing, and AI more broadly may facilitate or hinder progress toward Sustainable Development Goal targets, including those concerning inequality [10,11].

These contrasting possibilities raise questions about the distribution of opportunities to benefit from GenAI, as reported use varied across student groups and institutional settings, while perceived acceptability and educational benefits showed variable relationships with socioeconomic and demographic characteristics during an early stage of adoption [12,13]. Regulatory restrictions, subscription costs, language proficiency, and cultural fit can shape which tools are used for academic tasks [14]. At the same time, personalized support and feedback tailored to different linguistic needs have been proposed as ways to reduce barriers and broaden participation in learning [15]. Examining equity requires looking beyond equipment to the skills, values, and institutional support that may enable meaningful engagement [16]. This concern extends to AI in education more generally, where technological availability alone does not guarantee inclusion and the social conditions surrounding access to quality education also require attention [17].

Against this backdrop, the present systematic review examines how GenAI is used to individualize learning and provide student support in higher education, alongside favorable and unfavorable outcomes reported for creative agency and equitable learning. It also explores how the identified approaches and findings relate to SDG 4 and SDG 10 from the perspective of sustainable and inclusive higher education. Its contribution lies in integrating these educational dimensions and assessing the scope of the evidence in light of methodological quality and the distinction between perceptions and evaluated outcomes. In interpreting the implications for the SDGs, the review distinguishes connections explicitly stated in the studies from those developed through review-level interpretation of their findings, without assuming progress beyond what the evidence supports. Three research questions guide the review:

- RQ1. How does Generative AI enable learning individualization and student support in sustainable higher education?

- RQ2. What positive and negative outcomes related to creative agency and equitable learning emerge from studies on Generative AI-supported higher education?
- RQ3. How are the approaches and outcomes identified in the literature connected to the goals of SDG 4 and SDG 10?

2. Conceptual Framework

2.1. Generative AI and Pedagogical Integration in Higher Education

GenAI encompasses technologies capable of producing content, such as text or images, in response to different kinds of instructions through deep learning techniques [18]. Large language models with generative capabilities (LLMs) form part of this broader category, but generating responses or materials does not in itself constitute a pedagogical approach, whose design requires educational purposes, contextual knowledge, and human decisions about learning experiences [19].

Integrating applications based on these models also requires consideration of their practical feasibility, transparency, and ethical conditions through a human-centered approach [20]. In higher education, this perspective positions the tools as supporting resources within a process that brings together curriculum design, preparation for their use, and institutional responsibility, while preserving educator involvement and students' active participation [21].

2.2. Learning Individualization and Student Support

Individualization, personalization, and adaptive learning share an interest in responding to differences among students, although their terms and components are not consistently delineated [22]. Individualized teaching involves adjusting methods, activities, pacing, and support to each person's characteristics and needs, whereas personalized learning also incorporates their interests, goals, and participation in educational decisions [23]. In adaptive environments, changes to content or learning pathways are linked to information about the student's knowledge, performance, and other requirements [24]. Because these terms overlap, this review uses learning individualization as an umbrella term for adjustments to activities, resources, pacing, or support in response to particular needs.

These adjustments can be situated along a continuum, from predominantly learner-driven choices of goals and tasks to system-driven decisions informed by learning data; offering choices does not ensure that the most appropriate options are selected, and relevant learner characteristics may change during the process [25]. Within this conceptual delimitation, configuring a GenAI tool for a course does not, by itself, demonstrate adaptation of its guidance to information about each learner.

Student support encompasses assistance with academic demands through explanations, writing guidance, feedback on work, and opportunities for self-assessment [26]. Proposed applications of LLMs also include exercise generation, explanations of solutions, and assistance with programming and problem-solving, with human oversight [27]. Providing these functions does not necessarily individualize learning, since the availability of assistance is distinct from its adjustment to the recipient's characteristics, needs, or progress.

2.3. Creative Agency

Creative agency is understood here as students' capacity to participate actively in learning, make decisions, and retain ownership of the process, with a creative dimension that involves generating alternatives, evaluating their suitability, and choosing which ideas to develop when working with GenAI [28,29].

Receiving automated guidance and learning to regulate one's own work are distinct processes. In AI-assisted settings, improved performance with external support does not by itself demonstrate the capacity to act autonomously when that support is withdrawn [30]. Cognitive offloading offers a complementary perspective on concerns about overreliance. Laboratory evidence indicates that transferring part of cognitive processing to external resources can facilitate immediate task performance while affecting later retention; these consequences vary with the conditions of use, and an explicit intention to learn may attenuate adverse effects on memory [31].

2.4. Equitable Learning in GenAI-Supported Higher Education

Equitable learning is understood here as the opportunity to access, participate in, and benefit from GenAI-supported learning on fair terms. Digital divide research distinguishes between having access to a technology, possessing the capabilities to use it, and obtaining benefits from its use [32]. Inclusion also depends on how institutional expectations, permitted practices, and forms of communication allow students' needs and contributions to be recognized [33].

From this perspective, GenAI has been proposed as an accessibility resource for neurodivergent students who encounter barriers to written expression, offering support without replacing their intellectual contribution [34]. It has also been argued that generated content may reproduce dominant values and identities while de-emphasizing others, and that addressing the resulting equity risks requires institutional and sector-wide responsibility rather than falling solely on individual educators [35].

2.5. SDG 4 and SDG 10 as an Interpretive Lens for Sustainable Higher Education

Sustainable higher education is conceptualized as an institutional process connecting teaching, inclusion, and support for people with resource management and economic sustainability, with attention to the relationships among these dimensions [36]. This review focuses on its educational and social components, particularly opportunities to learn, participate, and receive support through GenAI.

Considering SDG 4 alongside SDG 10 has been proposed as a way to connect educational quality with equity and examine barriers to equity-focused quality in higher education [37]. To delimit this perspective, the review considers targets 4.4 on relevant skills, 4.5 on equitable access and reduced educational disparities, 4.7 on knowledge and skills for sustainable development, 10.2 on inclusion, and 10.3 on equal opportunities. Within universities, SDG 4 targets have been related to teaching and learning, whereas targets 10.2 and 10.3 have been linked to institutional governance and equity [38]. These connections require cautious interpretation, since commonly used indicators may offer only a partial view of quality and equity, and the initiatives described are not always accompanied by systematic evaluation of their outcomes [39].

Indicator 4.7.1 concerns the incorporation of global citizenship education and education for sustainable development into policies, curricula, educator training, and student assessment; broader monitoring has been proposed to include educational processes and outcomes [40]. In this review, the SDGs guide the interpretation of findings, distinguishing connections explicitly stated in the studies from those developed from their reported findings, without equating thematic alignment with demonstrated progress. The four constructs of the framework are understood as interrelated processes. Learning individualization and student support address responsiveness to students' needs; creative agency encompasses their capacity to make decisions, generate ideas, and retain ownership of the process; and equitable learning concerns the conditions that enable access, participation, and benefit.

3. Materials and Methods

3.1. Review Design, Protocol, and Reporting Framework

This systematic review examined empirical evidence on the use of generative artificial intelligence (GenAI) for learning individualization and student support in higher education. Particular attention was given to creative agency and equitable learning, as well as to the implications of these areas for sustainable and inclusive education. The scope of the review was defined using the Population–Interest–Context (PICo) framework. The population comprised higher education students; the phenomenon of interest concerned the use of GenAI to individualize learning and provide student support; and the context was sustainable higher education. The three research questions established in the protocol remained unchanged throughout the review.

A protocol was developed prospectively and registered with the International Platform of Registered Systematic Review and Meta-analysis Protocols (INPLASY) on 4 June 2026 (INPLASY202660012; https://doi.org/10.37766/inplasy2026.6.0012). The protocol was prepared in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) [41], while the completed review was reported following the PRISMA 2020 statement [42]. It specified the research questions, eligibility criteria, information sources, study-selection and data-extraction procedures, and the planned approach to evidence synthesis.

During full-text assessment, the outcome-related eligibility criterion was narrowed relative to the registered protocol. Although the original criterion allowed a study to qualify solely through sustainability-related educational implications, the decision rule applied in the review required each eligible study to provide empirical evidence relevant to at least one outcome dimension under RQ1 and at least one outcome dimension under RQ2. This restriction was introduced to prevent the inclusion of studies that referred to sustainability or the Sustainable Development Goals without examining the educational phenomena at the core of the review, namely learning individualization or student support, together with creative agency or equitable learning. RQ3 was consequently treated as an interpretive dimension through which the sustainability implications of the eligible evidence were examined, rather than as an independent route to study inclusion.

This post-registration change was introduced during full-text assessment and was not entered as a formal amendment in the INPLASY record. It did not modify the research questions, target population, higher education context, focus on GenAI, publication restrictions, or overall synthesis framework. The eligibility rule ultimately applied and its methodological rationale are described in Section 3.3. The agreement analysis was extended beyond the protocol-specified Cohen's kappa by also reporting Gwet's AC1 and bootstrap confidence intervals, as described in Section 3.6.

3.2. Information Sources and Search Strategy

The electronic search was conducted across four prespecified information sources: Scopus, ERIC, ScienceDirect, and the MDPI Journals Platform. Scopus was selected for its broad multidisciplinary coverage of higher education, artificial intelligence, and educational technology. ERIC provided specialized access to education research, whereas ScienceDirect and the MDPI Journals Platform enabled targeted retrieval from publisher-based journal collections covering education, technology, artificial intelligence, and sustainability. ScienceDirect and MDPI were therefore treated as publisher search platforms rather than comprehensive bibliographic databases, and some overlap with Scopus was anticipated.

All searches were executed, and the resulting records were exported, on 27 August 2026. The four sources differed in their search interfaces and available filters. In Scopus, the equation was applied to titles, abstracts, and keywords through the TITLE-ABS-KEY field.

In ERIC, it was entered in the basic search box under the "Collection" tab, without applying a field-specific restriction. The ScienceDirect equation was entered in the Advanced Search field "Find articles with these terms", while the MDPI equation was entered in the "Title/Keyword" search field. Search syntax, field codes, truncation, and equation length were adapted to these operational differences.

The search strategies combined three conceptual components: GenAI, higher education, and educational processes or outcomes. The GenAI component included terms such as "generative AI", "generative artificial intelligence", ChatGPT, "large language model", "AI tutor", and "conversational AI". The higher education component included "higher education", universit*, college*, and "tertiary education". Synonymous or related terms within each component were connected using OR, whereas the three components were combined using AND. The terms actually included varied across sources. Scopus included outcome terms for learning individualization, student support, creative agency, equitable learning, inclusion, accessibility, and sustainability. ERIC covered several of these areas but omitted some terms included in Scopus. In ScienceDirect and the MDPI Journals Platform, the outcome block was restricted to "creative agency" or "creativity." These differences concerned substantive search coverage as well as syntax and implementation.

The complete equations and filters actually used are presented in Table 1. A shorter equation was used in ScienceDirect because of platform restrictions on Boolean connectors, and the same equation was applied to the MDPI Journals Platform. Their narrower outcome vocabulary may have missed studies addressing individualization, student support, or equity without creativity-related terminology. The broader searches in Scopus and ERIC and complementary citation searching provided additional identification routes, but did not establish equivalent coverage across sources or eliminate this risk. No supplementary searches using expanded outcome terms were conducted. The magnitude of any resulting omissions could not be quantified, and this limitation is considered in Section 5.4.

Table 1. Search strategies and filters applied across the information sources.

Information Source	Search Equation	Filters Applied
Scopus	TITLE-ABS-KEY ("generative AI" OR "generative artificial intelligence" OR ChatGPT OR "large language model*" OR LLM OR "AI tutor*" OR "conversational AI") AND TITLE-ABS-KEY ("higher education" OR universit* OR college* OR "tertiary education") AND TITLE-ABS-KEY ("personalized learning" OR "learning individualization" OR "student support" OR "creative agency" OR "equitable learning" OR inclusion OR accessibility OR sustainability)	Publication year: 2022–2026; document type: Article; source type: Journal; language: English
ERIC	("generative AI" OR "generative artificial intelligence" OR ChatGPT OR "large language model*" OR "conversational AI") AND ("higher education" OR universit*) AND ("personalized learning" OR "student support" OR "creative agency" OR inclusion OR accessibility OR sustainability)	Publication date: Since 2022 (last 5 years); publication type: Journal Articles; peer-reviewed only
ScienceDirect	("generative AI" OR ChatGPT) AND ("higher education" OR university) AND ("creative agency" OR creativity)	Publication year: 2022–2026; article type: Research articles
MDPI Journals Platform	("generative AI" OR ChatGPT) AND ("higher education" OR university) AND ("creative agency" OR creativity)	Publication year: 2022–2026; article type: Article

At the time of the search, the ERIC "Since 2022" filter covered publications from 2022 through 2026. ERIC was the only source in which peer-reviewed status was restricted through a dedicated native filter. For records retrieved from Scopus, ScienceDirect, and

the MDPI Journals Platform, peer-reviewed status was verified during screening. The English-language restriction was applied directly in Scopus and verified during screening for records retrieved from the other three sources. Platform filters supported record retrieval but did not replace the application of the complete eligibility criteria during study selection.

Records exported from the four sources were consolidated for duplicate removal and subsequent screening. Backward and forward citation searching was undertaken as a complementary identification procedure and is described separately in Section 3.4.

3.3. Eligibility Criteria

Eligibility was assessed across five domains: population and educational context, focus on generative artificial intelligence, relevance of the educational outcomes, empirical study design, and publication characteristics. A study had to satisfy the requirements in all five domains to be included. Mixed populations or educational levels were considered only when the evidence corresponding to higher education students could be identified and interpreted separately.

The outcome criterion applied during full-text assessment was narrower than initially stated in the registered protocol. The protocol allowed eligibility when a study reported at least one outcome related to learning individualization, student support, creative agency, equitable learning, or sustainability-related educational implications. In the completed review, eligible studies were required to provide empirical evidence for at least one dimension of RQ1—learning individualization or student support—and at least one dimension of RQ2—creative agency or equitable learning. Evidence related solely to sustainability, SDG 4, or SDG 10 did not provide an independent basis for inclusion. These connections were examined under RQ3 only after a study had met the central educational outcome requirement.

Each report excluded after full-text assessment was assigned one principal reason and counted once in the PRISMA flow diagram. When more than one exclusion criterion applied, the recorded reason did not imply that the report satisfied the remaining criteria. The five primary exclusion labels were retained from the protocol. Within "Not relevant educational outcomes," the eligibility clarification distinguished four groups: "RQ1 only," "RQ2 only," "Sustainability or SDGs only," and "No relevant outcome." This subdivision distinguished reports addressing a single review-question cluster or sustainability alone from those providing no relevant outcome evidence, without changing the primary exclusion label. Table 2 presents the eligibility criteria and primary exclusion labels; Supplementary Material File S1 provides the report-level decisions and clarifications.

Table 2. Eligibility criteria and corresponding full-text exclusion labels.

Criterion	Inclusion Criteria	Exclusion Criteria
Population and educational context	Empirical studies involving students enrolled in universities or other tertiary-level institutions. Pre-service teachers were eligible only when they were enrolled in a higher education program and participated in the study in their role as students. Studies with mixed populations or educational levels were eligible only when data concerning higher education students could be separated from those of other participants.	**PRISMA label: Not higher education context.** Studies conducted in primary or secondary education, corporate training, professional development, non-formal education, or other non-university settings. Faculty-only or administrator-only studies were excluded, as were mixed-sample studies in which evidence concerning higher education students could not be examined separately.

Table 2. *Cont.*

Criterion	Inclusion Criteria	Exclusion Criteria
Generative AI focus	Studies explicitly examining an educational use of GenAI tools or systems, including ChatGPT, large language models, conversational GenAI, AI tutors powered by generative models, or multimodal GenAI applications used in teaching and learning.	**PRISMA label: Not generative AI focus.** Studies centered on conventional artificial intelligence, predictive analytics, automated classification, non-generative tutoring systems, or general discussions of AI without an identifiable educational application of GenAI.
Relevant educational outcomes	Studies providing empirical evidence for at least one RQ1 dimension—learning individualization or student support—and at least one RQ2 dimension—creative agency or equitable learning. Related constructs such as autonomy, idea generation, problem-solving, accessibility, inclusion, digital equity, or unequal access were considered when they clearly represented one of these dimensions.	**PRISMA label: Not relevant educational outcomes.** Studies that did not provide evidence relevant to both the RQ1 and RQ2 outcome structures. This category included studies limited to acceptance, readiness, academic integrity, technical performance, general perceptions, or sustainability implications when their findings could not be mapped to the required educational dimensions. Studies addressing only RQ3 were also excluded.
Empirical study design	Original empirical journal articles using qualitative, quantitative, or mixed-methods designs, including descriptive, case-study, survey-based, experimental, and quasi-experimental research.	**PRISMA label: Ineligible empirical design.** Reviews, editorials, commentaries, letters, book chapters, conference papers, preprints, theoretical essays, opinion papers, or other publication types that did not provide eligible original empirical evidence.
Publication characteristics	Peer-reviewed publications written in English, published between 2022 and 2026, available in full text, and compliant with the predefined publication filters.	**PRISMA label: Ineligible publication characteristics.** Reports published outside the predefined period, written in a language other than English, lacking peer-reviewed status, or otherwise failing to meet the predefined publication filters.

3.4. Study Selection Process and Citation Searching

Records retrieved from the four information sources were first imported into Zotero, version 10.0.2 (Digital Scholar, Falls Church, VA, USA). Potential duplicates were checked and removed before the remaining library was transferred to Rayyan (https://www.rayyan.ai/ (accessed on 4 November 2026) web-based application; Rayyan Systems Inc., Cambridge, MA, USA) for a second duplicate check, screening management, and recording of eligibility decisions. References flagged in either application were verified before removal to avoid deleting distinct publications with similar bibliographic information. No automated system was used to determine study inclusion or exclusion.

Two reviewers independently screened the titles and abstracts against the criteria described in Section 3.3. Records that appeared potentially eligible, or for which the available information was insufficient to support a decision, were retained for full-text assessment. The same reviewers then evaluated the retrieved reports independently. Disagreements were resolved through discussion and, when consensus could not be reached, consultation with a third reviewer. Each report excluded at this stage was assigned one principal reason using the five categories presented in Table 2.

The full texts were also examined for indications that multiple publications originated from the same underlying empirical investigation. When linked publications contributed complementary evidence, they were retained as separate reports under a shared study identifier. Report-specific identifiers preserved the provenance of the extracted information, while study-level counting prevented the same participants, intervention, or investigation from being represented more than once.

After the initial full-text assessment, backward citation searching was conducted by manually examining the reference lists of the eligible reports. Forward citation searching was then performed using the Scopus "Cited by" function to identify subsequent publications that cited each eligible report. The included literature served as the seed set for both procedures. Backward and forward citation searching was completed on 18 September 2026.

Records located through citation searching were compared with the existing reference library and checked for duplication. New candidates underwent title and abstract screening and, where relevant, full-text assessment using the same eligibility criteria applied to records retrieved from the original information sources. Their identification route was recorded in the extraction matrix as original information-source search, backward citation searching, or forward citation searching. This maintained a traceable connection between the route of discovery and the final evidence base. The numerical results of the selection process are reported in Section 4.1 and summarized in the PRISMA flow diagram.

3.5. Data Extraction and Evidence Synthesis

Data extraction and synthesis were organized to preserve the connection between each publication, its underlying study, and the research questions. Information was extracted at the report level, whereas evidence was combined and summarized at the study level. This distinction prevented linked publications arising from the same investigation from being counted as independent studies.

3.5.1. Data Extraction Process

Data were extracted manually from the full-text reports and any accompanying supplementary materials using a standardized Microsoft Excel 2021 (Microsoft Corporation, Redmond, WA, USA) matrix developed for this review. Initial data extraction was conducted between 21 and 25 September 2026, as scheduled in the registered protocol. Subsequent extraction used the same procedures. One reviewer extracted the data from each report into the shared matrix, and a second reviewer checked the completed extraction against the source publication. Decisions requiring interpretation were resolved through discussion and consensus.

Each investigation received a unique Study ID. When two publications reported findings from the same underlying investigation, they were assigned separate Report IDs but retained a shared Study ID. The "Role in synthesis" field indicated whether a report provided the sole evidence for a study or contributed complementary findings alongside another publication. This structure allowed information from linked reports to be retained without duplicating the study in the descriptive counts or evidence synthesis.

No information was inferred or imputed when a report did not provide sufficient evidence. A dimension that was not examined or reported was recorded as "not reported." For RQ3, "not supported" indicated that the available findings did not justify a connection with the corresponding Sustainable Development Goal. Study authors were not contacted for additional information because the synthesis was based on the evidence available in the published reports and their supplementary materials. The completed extraction matrix is provided as Supplementary Material File S2.

3.5.2. Data Items

The extraction matrix captured bibliographic, methodological, contextual, and outcome-related information. Descriptive fields included the Study ID, Report ID, complete reference, role in the synthesis, report title, identification source, country, population and educational context, sample characteristics, study design, and data-collection methods or measures.

Evidence relevant to RQ1 was extracted separately for learning individualization and student support. The first dimension covered findings concerning personalization,

adaptive learning, differentiated learning pathways, feedback tailored to individual needs, and learner control over the use of GenAI, which was distinguished from system adaptation as described in Section 3.5.3. The second encompassed academic guidance, scaffolding, feedback, tutoring, emotional or motivational assistance, accessibility, and other forms of support enabled by generative artificial intelligence.

For RQ2, creative agency and equitable learning were coded as distinct dimensions. Creative agency included evidence related to idea generation, authorship, creative expression, experimentation, problem-solving, and students' capacity to exercise control over AI-assisted learning activities. Equitable learning covered accessibility, inclusion, participation, differences in access or use, support for diverse learners, and the risk that generative AI might reproduce or intensify existing educational inequalities.

RQ3 was represented by two fields addressing implications for SDG 4 and SDG 10. Statements explicitly connecting the findings with quality education, inclusion, accessibility, or the reduction in inequalities were distinguished from connections developed through review-level interpretation. Such interpretations were made cautiously and only when supported by the study's empirical findings; they were not treated as evidence that the original authors had claimed a direct contribution to either Sustainable Development Goal.

The original RQ1 and RQ2 extraction fields contained structured narrative summaries rather than numerical scores. Descriptors such as 'direct,' 'positive,' 'mixed,' 'strong,' 'limited,' 'perception-based,' and 'self-reported' were used flexibly within those summaries; they were distinct from the standardized thematic and direction codes subsequently applied for synthesis, as described in Section 3.5.3. For RQ3, the matrix distinguished among "Direct," "Indirect—review-level interpretation," and "Not supported," with each designation followed by a study-specific explanation.

3.5.3. Evidence Synthesis

The evidence was synthesized using descriptive summaries and a narrative synthesis. Study characteristics were summarized by publication year, country or geographical setting, participant population, sample size, research design, and route of identification. Frequencies and percentages were used where they provided an informative overview. When an investigation was represented by more than one report, findings were integrated under the shared Study ID before study-level counts were calculated.

For RQ1, evidence was coded by report into non-mutually exclusive categories of individualization approach and support function. Individualization categories comprised learner-directed individualization, purpose-built or course-customized systems, system-driven adaptive personalization, and educator-designed differentiation. Learner-directed individualization described students' control over the timing, focus, prompts, or pace of GenAI use; it did not establish that the system diagnosed learning needs or adapted a learning pathway. Purpose-built or course-customized systems were developed or configured for the study or learning context. System-driven adaptive personalization required documented adjustments to guidance based on learner profiles, performance, or developing responses. Support functions were coded separately, because receiving explanations, feedback, or other assistance did not by itself establish individualization.

For RQ2, findings on creative agency and equitable learning were coded separately by report into non-mutually exclusive favorable and unfavorable themes. Each dimension received one direction category: positive, negative, mixed, no clear directional finding, or not reported. Direction reflected findings within that dimension, including participants' experiences and perceptions, rather than methodological limitations or concerns proposed only by the study authors. A mixed classification required both favorable and unfavorable findings, but did not imply that they were equally frequent, equally strong, or measured

using comparable methods. Null or solely descriptive findings were not classified as negative. Interpretations distinguished assessed performance from perceived benefits, reported practices, and anticipated risks; perceived creativity, autonomy, or critical thinking was not treated as demonstrated improvement in those outcomes.

Equitable-learning themes required evidence concerning an identifiable learner group or a specific barrier to equitable access or participation. General convenience or on-demand availability alone did not qualify. Descriptive access-related findings without a supported favorable or unfavorable direction were coded as having no clear directional finding and were not interpreted as demonstrated reductions in educational disparities. Access to academic assistance could support the student-support dimension of RQ1, but did not independently establish equitable learning under RQ2. A study could nevertheless satisfy RQ2 through creative-agency findings. Findings from all GenAI-supported conditions and, where applicable, linked reports were considered before assigning the study-level direction. The report-level coding is recorded in Supplementary Material File S2.

For RQ3, references made by study authors to an SDG, a specific target, or an official indicator were distinguished from connections interpreted at the review level. Connections were classified as "Direct," "Indirect—review-level interpretation," or "Not supported," with study-specific explanations; advocacy for equitable access or free tool availability alone did not establish an SDG 10 connection. For each SDG, studies were classified by their highest documented level of connection. Reported findings from RQ1 and RQ2 were then related descriptively to SDG targets 4.4, 4.5, 4.7, 10.2, and 10.3. This mapping did not assess progress toward any target. Target 4.3 was not included, limiting the mapping's coverage of equal access to tertiary education.

Exploratory study-level cross-tabulations compared RQ2 directions and unfavorable themes by type of GenAI use, focus on learners at risk of educational disadvantage, region, disciplinary area, and presence of a purpose-built or course-customized system. The relevant classifications are recorded by report in Supplementary Material File S2 and were consolidated for study-level comparisons. Educational modality was not examined because it was inconsistently reported. The subgroup comparisons were descriptive; no statistical tests were performed.

The narrative synthesis was organized around six dimensions: learning individualization, student support, creative agency, equitable learning, implications for SDG 4, and implications for SDG 10. Within each dimension, it examined recurring findings, convergence and disagreement, contextual variation, limitations of the evidence, and insufficiently investigated topics. Interpretations reflected the underlying study designs and measures, with particular caution for findings based mainly on perceptions or self-reported experiences. A meta-analysis was not undertaken because studies differed substantially in design, population, educational setting, GenAI use, and outcome measurement. Methodological appraisal informed interpretation but was not used to assign numerical study weights or determine eligibility. The appraisal and sensitivity-analysis procedures are described in Sections 3.6 and 3.7, respectively.

Python (version 3.13.16) was used in Google Colab (Google LLC, Mountain View, CA, USA; accessed October 2026) for statistical calculations and figure preparation. ChatGPT (GPT-6.1 Sol; OpenAI, San Francisco, CA, USA) assisted with developing and revising Python code for statistical calculations and plotting; the authors reviewed the code and verified the outputs against the extracted data. ChatGPT's image-generation feature was used to develop the graphical abstract, which was reviewed and approved by the authors.

3.6. Methodological Quality Appraisal Using the Mixed Methods Appraisal Tool

The methodological quality of the studies included in the analysis and evidence synthesis was appraised using the 2018 version of the Mixed Methods Appraisal Tool (MMAT) [43]. The MMAT was selected because it provides a common framework for critically examining qualitative research, randomized controlled trials, non-randomized studies, quantitative descriptive studies, and mixed methods research. Its design-specific structure was suitable for the methodological diversity of the evidence included in this review.

The two MMAT screening questions were applied to each study before its methodological design was classified. The five criteria for the assigned design category were then assessed. For mixed methods studies, the relevant qualitative and quantitative components were appraised alongside the five mixed methods criteria addressing the rationale for the design, integration of components, interpretation of integrated findings, treatment of divergences, and adherence of each component to its methodological tradition. When an underlying study was represented by more than one report, the linked publications were considered jointly and received a single study-level appraisal.

Before the full appraisal, the two reviewers piloted the procedure independently on a purposive subset representing the principal study designs found in the review. Differences in interpretation were discussed to refine the review-specific decision rules without modifying the original MMAT criteria. The resulting appraisal framework, response conventions, and operational guidance are presented in Supplementary Material File S3.

The two reviewers subsequently evaluated each study included in the evidence synthesis independently. Each applicable criterion was classified as "Yes," "No," or "Can't tell," in accordance with the MMAT guidance. Brief justifications were recorded for "No" and 'Can't tell' judgments, with the location of supporting information indicated where available. The initial appraisal matrices were retained unchanged until both independent assessments had been completed. Disagreements were then examined through discussion and resolved by consensus; a third reviewer was consulted when agreement could not be reached. The independent ratings, criterion-level comparisons, discrepancies, and final consensus decisions are documented in Supplementary Material File S4; the statistical summaries of inter-rater agreement are provided separately in Supplementary Material File S5.

Inter-rater agreement was calculated from the two reviewers' initial, independent item-level judgments, before consensus, using the response categories "Yes," "No," and "Can't tell." Only criteria applicable to each study were included. Overall agreement was examined both with and without MMAT screening items S1 and S2; agreement was also summarized by response category and MMAT criterion block. The analyses included observed percentage agreement, unweighted Cohen's kappa, and Gwet's AC1. Ninety-five percent confidence intervals for overall percentage agreement, Cohen's kappa, and Gwet's AC1 were estimated from 5000 bootstrap resamples, using the study as the resampling unit and the percentile method. Cohen's kappa was interpreted using conventional agreement ranges [44] and considered alongside observed agreement and Gwet's AC1 because its value can be affected by the distribution of response categories. The paired judgments are available in Supplementary Material File S4, the agreement statistics in Supplementary Material File S5, and the principal findings are reported in Section 4.3.1.

Consistent with the MMAT recommendations, criterion-level judgments were not combined into a numerical total score, and the studies were not assigned overall categories such as high, moderate, or low quality. No study was excluded solely because of its appraisal findings. Instead, methodological strengths and limitations were retained at the domain level and used to qualify the interpretation of the synthesized evidence. Their influence on the stability of the review conclusions was examined through the sensitivity procedures described in Section 3.7.

3.7. Sensitivity Analysis Procedure

After the MMAT consensus judgments were finalized, an exploratory, study-level sensitivity analysis compared the primary narrative synthesis with a more restrictive synthesis. The comparison examined whether the principal patterns and interpretations across the six review dimensions remained supported when studies meeting the specified criterion-level omission rule were temporarily set aside.

Studies were not selected for sensitivity analysis according to a numerical MMAT score or a predetermined number of unfavorable responses. Instead, the assessment focused on criterion-level judgments. For the restricted sensitivity comparison, a study was temporarily omitted from every synthesis dimension to which it contributed when the final MMAT consensus included a 'No' judgment in a prespecified criterion directly related to the derivation or substantiation of findings, group comparability, outcome completeness, measurement, confounding, intervention fidelity, statistical analysis, or mixed-methods integration and interpretation (1.3–1.5, 2.2–2.5, 3.2–3.5, 4.3–4.5, or 5.2–5.4). A 'Can't tell' judgment alone did not trigger omission; neither did limitations limited to sampling or representativeness, insufficient reporting of random-sequence generation under criterion 2.1, or criterion 5.5 alone. The criterion-level basis and rationale for each temporary omission are documented in Supplementary Material File S6.

Criterion 2.1 did not independently trigger omission because incomplete reporting of sequence generation did not establish flawed randomization; criteria 2.2–2.5 addressed related limitations in trial conduct and outcomes. Criterion 5.5 summarizes component-level methodological quality; omission was determined by the specified component and integration criteria rather than this aggregate judgment alone. Limitations under both criteria remained relevant to interpretation even when they did not trigger omission.

For each of the six review dimensions—learning individualization, student support, creative agency, equitable learning, implications for SDG 4, and implications for SDG 10—the primary synthesis was compared with a restricted synthesis retaining only studies that did not meet the temporary-omission rule. The comparison considered the direction and recurrence of the principal findings, the breadth of supporting evidence, contextual variation, and whether an interpretation depended substantially on studies with identified methodological limitations. For the SDG-related dimensions, explicit statements in the included studies were distinguished from review-level interpretations, using target descriptions drawn from the United Nations 2030 Agenda [45].

These analyses were descriptive comparisons rather than statistical tests of robustness. Persistence of a finding in the restricted synthesis indicated that it remained represented among the retained studies; it did not demonstrate an unchanged effect magnitude, eliminate bias, or establish high confidence in the finding. Interpretation also considered the reduced breadth and composition of the supporting evidence. Temporary omission applied only to the sensitivity comparison; all included studies remained represented in the primary synthesis. The study-level decisions, dimension-level comparisons, and their effects on interpretation are documented in Supplementary Material File S6; the findings are reported in Section 4.8.

4. Results

4.1. Study Selection and Citation Searching

The searches across the four electronic information sources identified 3405 records: 772 from Scopus, 37 from ERIC, 2494 from ScienceDirect, and 102 from the MDPI Journals Platform. After removing 39 duplicates, 3366 unique records remained for title-and-abstract screening. No records were removed through automated procedures or for reasons unrelated to duplication. At the initial screening stage, 3154 records were excluded because

they did not meet the eligibility criteria. The remaining 212 reports were sought for retrieval through the searched information sources and publisher platforms. Two remained unavailable, leaving 210 reports for full-text eligibility assessment.

Full-text assessment resulted in the exclusion of 150 reports. The primary reasons were "Not relevant educational outcomes" (n = 122), "Not generative AI focus" (n = 11), "Not higher education context" (n = 9), "Ineligible empirical design" (n = 7), and "Ineligible publication characteristics" (n = 1). The information-source searches yielded 60 eligible reports representing 58 distinct studies. Two studies were each represented by two linked reports providing complementary findings. Backward citation searching identified three additional eligible reports, each corresponding to a separate study; forward citation searching produced no further inclusions. Intermediate counts of records identified, screened, and excluded through citation searching were not recorded; only the additional eligible reports were documented. The final evidence base comprised 61 studies represented by 63 reports. Figure 1 summarizes identification, screening, retrieval, eligibility assessment, and inclusion.

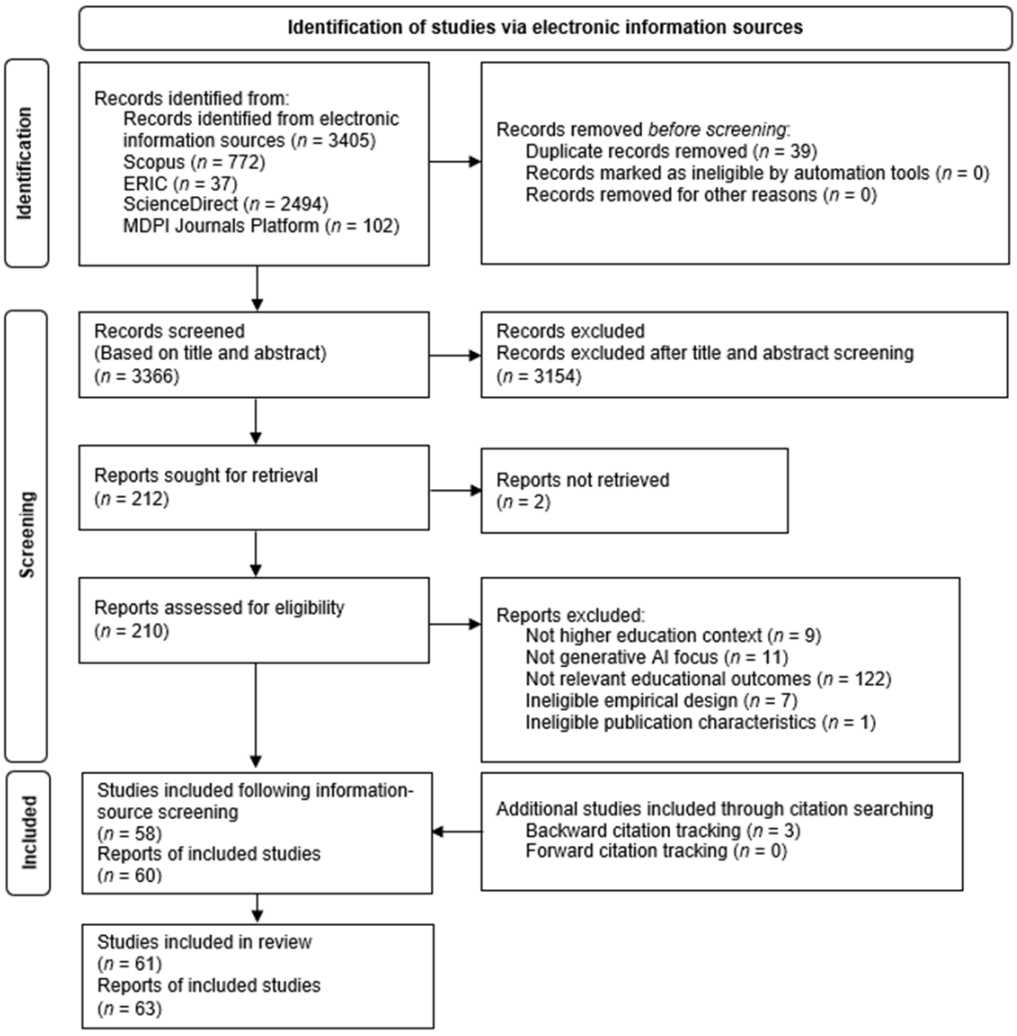


Figure 1. Study identification, screening, eligibility assessment, and final inclusion, including additional studies identified through citation searching. Note: For citation searching, only additional inclusions are shown because intermediate identification and selection counts were not recorded.

Within the outcome-related exclusions, 58 reports addressed RQ1 only, 17 addressed RQ2 only, six concerned sustainability or SDGs only, and 41 provided no relevant outcome evidence. Seventy-five reports therefore addressed only one of the two required educational

outcome clusters; including the six sustainability-only reports, 81 addressed a single outcome route potentially relevant under the broader registered criterion but did not meet the joint RQ1–RQ2 requirement. Because one principal exclusion reason was recorded per report, these figures do not establish how many would otherwise have satisfied all remaining eligibility criteria. The excluded reports, their primary reasons and eligibility clarifications, and the two reports not retrieved are documented in Supplementary Material File S1. The completed PRISMA 2020 checklist is provided in Supplementary Material File S7.

4.2. Characteristics of the Included Studies

The 61 included studies, represented by 63 reports [46–108], were published in 2026 (n = 33), 2025 (n = 25), 2024 (n = 2), and 2023 (n = 1). Reported study locations covered 37 countries or territories. Studies were grouped under Asia (n = 24), Europe (n = 18), North America (n = 5), Latin America and the Caribbean (n = 5), Africa (n = 2), and Oceania (n = 2); two were multiregional, and three did not explicitly report their location. Five studies involved more than one country. Türkiye was grouped under Europe, Israel under Asia, and Australia under Oceania.

Disciplinary groupings comprised multidisciplinary and other contexts (n = 22), STEM and computing (n = 14), language learning and writing (n = 13), health sciences (n = 7), and teacher education (n = 5). Participants included undergraduate and postgraduate students, with nine studies focused on learners at risk of educational disadvantage. Samples ranged from two focal cases to 1104 survey respondents. The evidence combined accounts of naturally occurring GenAI use with structured educational activities; student perceptions were common, while some studies assessed performance through tests, grades, or rubric-scored work. Table 3 summarizes study characteristics.

Table 3. Characteristics of the 61 included studies, represented by 63 reports.

Reference(s)	Country	Population and Sample	Study Design
[46]	The Netherlands	Bachelor's and master's students; 184 student survey responses (13% response rate)	Exploratory mixed-methods
[47]	United States	Neurodiverse STEM graduate students; N = 17	Qualitative interpretivist
[48]	China	Female undergraduate English-language learners; N = 2 focal cases from an initial survey of 867	Qualitative comparative multiple-case
[49]	Spain and Latin America	Online cybersecurity master's students; N = 350	Single-group mixed methods pretest–posttest
[50]	China	Educational Technology undergraduates; N = 94	Mixed methods study with individual random assignment
[51]	Spain	Undergraduate and graduate students across disciplines mainly engineering; N = 254	Mixed methods non-randomized comparison
[52]	Bangladesh	Undergraduates with learning disabilities or special learning needs; N = 28	Exploratory cross-sectional mixed methods
[53]	Saudi Arabia	First-year undergraduate research-writing students; N = 231 enrollments across three cohorts	Mixed methods comparative cohort study
[54]	Mexico	Undergraduate STEM engineering students; N = 28	Qualitative phenomenological
[55]	South Africa	Undergraduates from two public universities; N = 248 reported; demographic totals = 247	Explanatory sequential mixed methods
[56]	India	Pre-service science teachers enrolled as university students; N = 25	Single-group mixed methods pre–post pilot

Table 3. *Cont.*

Reference(s)	Country	Population and Sample	Study Design
[57]	Hungary	First-year university programming students; N = 120	Single-group mixed methods pilot
[58]	China	Undergraduate visual communication design students; N = 149	Mixed methods comparison of two GenAI tools across intact classes
[59]	Türkiye	Undergraduate preparatory-school English students; N = 86 assigned; 84 with valid proficiency scores	Three-group pretest–posttest with intact classes randomly assigned to conditions
[60]	Ecuador	Undergraduate agro-environmental engineering students; N = 1104 valid respondents	Quantitative cross-sectional survey
[61]	Republic of Korea	Undergraduate, master's, and doctoral students; N = 350 valid respondents	Quantitative cross-sectional correlational survey
[62]	India	Undergraduate and postgraduate English-language learners; N = 150; interview subset of 30	Single-group mixed methods pretest–posttest
[63]	United States	Higher education students with and without disabilities; N = 383 valid respondents	Exploratory cross-sectional survey with open-ended responses (mixed methods)
[64]	United States	Undergraduates in a sustainability and technology course; N = 39 students documenting AI use	Qualitative analysis of submissions and reflections
[65]	Nigeria, Kenya, and South Africa	University students in programming-related courses; N = 322	Cross-sectional quantitative comparative survey
[66]	Portugal	Students across higher education programs; N = 132 respondents; 129 GenAI users in most analyses	Cross-sectional quantitative survey
[67]	Switzerland, France, and Germany	Bachelor's and master's students with disabilities; N = 33	Exploratory qualitative interview study
[68]	United States	Undergraduate and graduate Geomatics students; N = 45	Quantitative observational classroom comparison
[69]	El Salvador	Undergraduates from three faculties; N = 365 survey respondents	Sequential mixed methods cross-sectional study
[70]	Antigua and Barbuda	Medical students (medical degree programme); N = 9	Qualitative study of a simulated clinical interaction
[71]	China	Undergraduate academic-English students; N = 310	Mixed methods quasi-experimental pretest–posttest
[72]	Indonesia	Undergraduate Mathematics Education students; N = 75	Single-group convergent mixed methods pretest–posttest
[73]	China	International university students; N = 100	Within-participant mixed methods quasi-experimental
[74]	Republic of Korea	Undergraduate occupational therapy students; N = 86	Randomized mixed–methods controlled pretest–posttest
[75]	Vietnam	Undergraduates with prior ChatGPT use; N = 263 valid survey responses; 10 interviewees	Cross-sectional mixed methods
[76]	Ukraine	Medical students, interns, and medical doctoral students; N = 244	Cross-sectional survey with quantitative and qualitative components

Table 3. *Cont.*

Reference(s)	Country	Population and Sample	Study Design
[77]	China	Undergraduate and graduate pre-service teachers; N = 48	Two-condition study with random assignment
[78,79]	United Kingdom	University students registered with disability support services; N = 124 valid responses from a shared survey reported in both reports	Exploratory cross-sectional survey with open-ended responses (mixed methods), reported in two linked reports
[80]	China	First-year university English-language learners; N = 67; interview subset of 10	Mixed methods study with two intact classes randomly designated to conditions
[81]	Indonesia	University students in an Economic English course; N = 50	Mixed methods quasi-experimental comparison
[82]	United Kingdom	Undergraduate Computer Science students; N = 21	Short-term longitudinal qualitative
[83,84]	Israel	Undergraduates in a Database Management course; N = 96 in the shared cohort; analytic samples varied by report	Mixed methods classroom investigation reported in two linked reports
[85]	China	Undergraduate and postgraduate students; N = 300	Longitudinal mixed methods configurational
[86]	United Kingdom	Taught and research postgraduate students; N = 23	Qualitative interviews and observation
[87]	Türkiye	Undergraduate Computer Science students; N = 45 with complete assessments	Randomized controlled pretest–posttest
[88]	China	Undergraduate and postgraduate EFL learners; N = 25 in the main Q-study; 17 preliminary interview participants reported separately	Exploratory Q-methodological study with post-sort interviews
[89]	United States; Thailand	International undergraduate and postgraduate students; N = 22: 14 in the United States and eight in Thailand	Qualitative multiple-case study
[90]	Hungary; Norway	Medical and dentistry students; N = 344 survey respondents; nine additional educator and physician interviewees	Cross-sectional, mixed-methods study
[91]	United Kingdom	Undergraduate and postgraduate students; N = 37 survey respondents; three specialist support-staff interviewees	Qualitative-priority convergent mixed-methods study and pedagogical framework development
[92]	China	First-year undergraduate EFL writing students; N = 66 in two intact classes; 11 interviewees	Mixed-methods quasi-experimental pretest–posttest study
[93]	Slovakia	Electrotechnics course students; N = 736 enrolments across four cohorts; within-year access comparison included 186 students	Retrospective observational study with historical cohort and non-randomized within-year comparisons
[94]	Australia	Undergraduate entrepreneurship students; N = 147 structured reflections from 176 enrolled students	Qualitative intervention-based case study
[95]	Australia	Undergraduate nursing students; N = 29 survey participants; 16 completed the entire survey; 87 completed the optional learning activity	Cross-sectional multi-method evaluation

Table 3. *Cont.*

Reference(s)	Country	Population and Sample	Study Design
[96]	India	University students across four professional disciplines; reported N = 19	Qualitative interview study with thematic and root-cause analyses
[97]	India	Third-year medical students; N = 253 analyzed, with 235 qualitative reflections	Mixed-methods, non-randomized delayed-intervention study
[98]	Not explicitly reported	Twenty undergraduate and 20 postgraduate students; N = 40	Qualitative descriptive study using email interviews and UTAUT-informed analysis
[99]	Türkiye	Social studies pre-service teachers; N = 20	Explanatory sequential mixed-methods study
[100]	Japan	Female undergraduates studying Chinese as a foreign language; N = 20	Explanatory sequential mixed-methods classroom case study over 14 weeks
[101]	United States	Students in a graduate-level GIS programming course; 24 consented, with item-specific samples	Mixed-methods
[102]	Not explicitly reported	Undergraduate and master's students across disciplines; 485 completed questionnaires	Quantitative cross-sectional survey
[103]	Taiwan	Undergraduates in a storytelling and scriptwriting course; N = 60	Twelve-week mixed-methods action research with single-group pre–post assessment
[104]	Not explicitly reported	First- and second-year design students; N = 40	Within-participant mixed-methods experiment with fixed independent and ChatGPT-assisted phases
[105]	Estonia	Programming students; 231 survey respondents from 323 enrolled students	Mixed-methods observational study
[106]	Germany	First-year international fashion design students; 25 participated and 21 were analyzed	Within-participant experiment with a fixed no-AI–AI–no-AI sequence
[107]	Colombia	Fourth-year engineering students; N = 128 analyzed from 151 initially enrolled	Randomized within-participant mixed-methods study
[108]	Spain	Final-year undergraduate nursing students preparing bachelor's theses; N = 66	Constructivist qualitative descriptive study

Note: N denotes the reported sample size; enrolled, responding, and analytic samples are distinguished where relevant. In the text, *n* denotes the number of studies. Reports [78,79] represent one study, as do reports [83,84]; each pair is combined in one row and counted once.

4.3. Methodological Quality Appraisal

4.3.1. Inter-Rater Agreement

The reviewers agreed on 661 of 797 independent paired MMAT judgments, yielding 82.9% agreement (95% CI: 79.4–86.4), Cohen's κ = 0.664 (95% CI: 0.593–0.731), and Gwet's AC1 = 0.772 (95% CI: 0.723–0.820). Both screening questions received concordant "Yes" ratings in all 61 studies; screening-only kappa was not estimable because responses did not vary. Excluding screening, agreement across 675 judgments was 79.9% (95% CI: 75.6–84.0), with κ = 0.642 (95% CI: 0.567–0.713) and AC1 = 0.720 (95% CI: 0.660–0.778). Both kappa point estimates fell within the conventional substantial-agreement range [44], although their confidence intervals extended into the moderate range.

Agreement within design-specific criterion blocks ranged from 77.3% for mixed methods criteria to 85.0% for quantitative descriptive criteria. Category-specific agreement was 89.0% for "Yes," 75.9% for "No," and 67.7% for "Can't tell." The 136 disagreements involved 45 studies and most frequently concerned integrated interpretation (5.3; *n* = 19),

qualitative data collection (1.2; n = 16), and substantiation of qualitative interpretations (1.4; n = 15). All were resolved through discussion without consulting a third reviewer. Supplementary Materials Files S4 and S5 provide the independent ratings, consensus decisions, and agreement statistics.

4.3.2. MMAT Appraisal Findings

All 61 studies met both MMAT screening questions. Of 675 applicable design-specific and component judgments, 377 were "Yes" (55.9%), 168 "Can't tell" (24.9%), and 130 "No" (19.3%). These were criterion-level assessments, not overall study-quality scores. Across 50 qualitative appraisals, criterion 1.4 received nine "No" judgments. Among 24 quantitative descriptive appraisals, representativeness (4.2) received 18 "No" judgments, while nonresponse bias (4.4) received 16 "Can't tell" and five "No" judgments. Confounding control (3.4) was rated "No" in all 16 non-randomized appraisals. Among eight randomized-trial appraisals, sequence generation (2.1) was unclear in six, while adherence (2.5) was adequate in all eight.

Among 37 mixed methods studies, integration (5.2) received 30 "Yes" judgments, but integrated interpretation (5.3) received 15 "No" judgments. Treatment of divergence (5.4) was unclear in 17 studies and inadequate in four. Criterion 5.5 received 33 "No" and four "Can't tell" judgments, indicating that full adherence to the methodological standards of both components was not established in any mixed methods study. These limitations qualified interpretation of the findings and informed the sensitivity analysis in Section 4.8; they did not determine exclusion from the primary synthesis. Detailed judgments are provided in Supplementary Material File S4.

4.4. Findings on Learning Individualization and Student Support (RQ1)

All 61 studies contributed evidence on student support, while 51 documented an individualization approach; ten reported assistances without an identifiable individualization approach [96–99,102–104,106–108]. Learner-directed control was coded in 46 studies (75.4%), purpose-built or course-customized systems in ten (16.4%), system-driven adaptive personalization in five (8.2%), and educator-designed differentiation in one (1.6%). Categories could overlap. Learner-directed control did not establish system adaptation, and these frequencies describe the included evidence rather than the wider literature. Table 4 summarizes the approaches and support functions.

Explanation and tutoring were reported in 52 studies (85.2%), followed by writing and task feedback in 40 (65.6%), self-regulatory and emotional support in 24 (39.3%), practice, assessment, and simulation in 20 (32.8%), and programming and problem-solving in 17 (27.9%). Assistance included conceptual clarification, draft revision, planning, conversational practice, and debugging. Self-regulatory and emotional support did not necessarily involve specialized emotional interventions.

These functions frequently co-occurred. The linked reports [83,84], for example, described immediate explanations, learner-controlled pacing, and comprehension questions. Figure 2A shows that explanation and tutoring accompanied all ten purpose-built or course-customized systems and all five studies reporting system-driven adaptive personalization. Of the 46 studies reporting learner-directed individualization, 41 also reported explanation and tutoring and 28 reported writing and task feedback. Figure 2B shows that eight studies reported one support function, 24 reported two, 19 reported three, and ten reported four; 53 studies (86.9%) reported at least two.

Table 4. Categories of learning individualization and student support reported in the included studies (n = 61).

Dimension	Category	Description	Studies, *n* (%)	References
Learning individualization	Learner-directed individualization	Students selected the timing, focus, prompts, or pace of GenAI use.	46 (75.4)	[54,82,88,100,101,105]
	Purpose-built or course-customized system	A GenAI system was developed or configured for the study or learning context.	10 (16.4)	[46,53,93]
	System-driven adaptive personalization	Documented guidance was adjusted using learner profiles, performance, or developing responses.	5 (8.2)	[49,50,62]
	Educator-designed differentiation	GenAI helped prospective teachers prepare materials or activities for different learner profiles.	1 (1.6)	[56]
Student support	Explanation and tutoring	On-demand clarification, explanations, summaries, or guidance on concepts and tasks.	52 (85.2)	[46,54,66,90,96–98,100–102,105–108]
	Writing and task feedback	Support for structuring, revising, or improving academic work.	40 (65.6)	[46,59,78,92,96,98–100,102–104,106–108]
	Practice, assessment, and simulation	Exercises, quizzes, conversational practice, or simulated interactions.	20 (32.8)	[46,62,70,88,97,100,102]
	Self-regulatory and emotional support	Planning, metacognitive prompts, management of learning difficulties, or emotional support.	24 (39.3)	[67,85,95,97,100,103,105,108]
	Programming and problem-solving	Assistance with code, debugging, or structured problem-solving tasks.	17 (27.9)	[46,65,82,93,101,102,104,105,107]

Note: Categories are not mutually exclusive. References are illustrative. Percentages use 61 studies as the denominator; ten reported no individualization approach. Linked reports were combined, and each study was counted once per category.

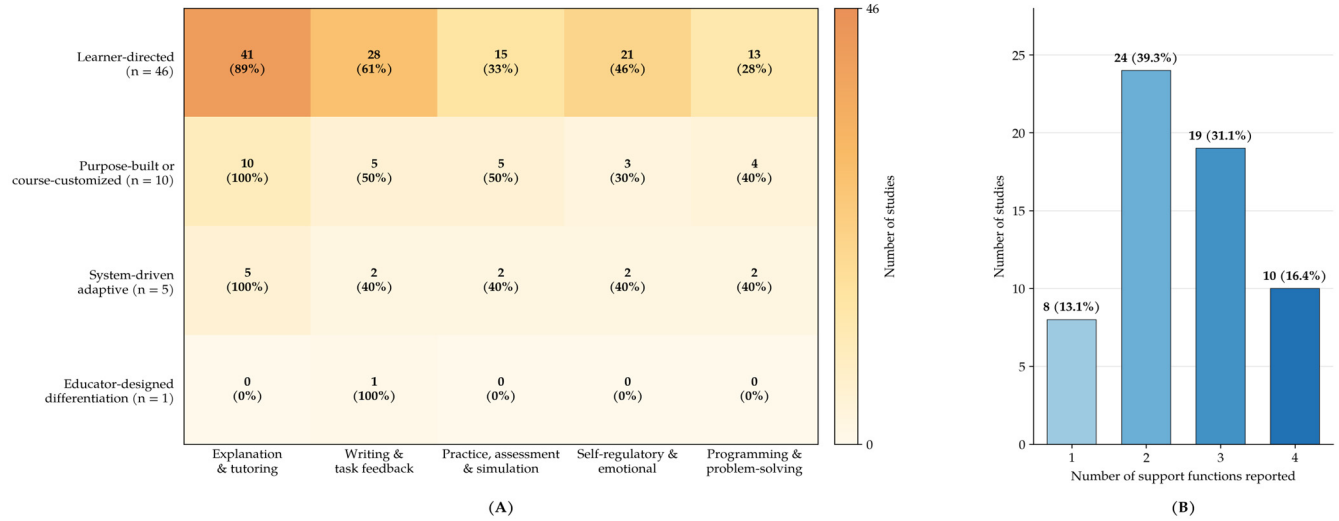


Figure 2. (**A**) Co-occurrence of learning individualization approaches and student support functions. Cell labels show the number of studies and the percentage within each individualization approach; color represents the number of studies. The ten studies that reported no individualization approach are documented in Supplementary Material File S2. (**B**) Number of support functions reported per study, with percentages calculated across 61 distinct studies. Linked reports were combined at the study level; categories in panel (**A**) can overlap.

The reported benefits were not uniform across studies or outcomes. In a retrieval-augmented system, 89% of students evaluated personalized content positively, and pre–post assessments showed improvements, although the absence of a comparison group prevented attribution of these changes to the system [49]. A Socratic conversational agent adjusted its guidance to students' developing responses, but the primary analysis did not establish a significant between-condition difference in conceptual knowledge [50]. Students using

a generative AI textbook described flexible pacing and accessible support, yet posttest achievement did not differ significantly from the comparison group [74]. A structured ChatGPT writing intervention supported planning and revision and showed gains in selected self-reported strategies, but did not establish improved writing proficiency [92]. In electrotechnics, official access to a course-customized assistant was associated with higher final-assessment success within one cohort; however, uncontrolled confounding and uncertainty about actual use limited causal interpretation [93].

Other studies documented perceived conceptual, linguistic, and reflective support without independently assessed learning gains. Examples included learner-selected language practice [88], academic assistance for international students [89], ChatGPT summaries for examination preparation [90], reflection on entrepreneurial cases [94], and AI-generated images used to examine disability-related stereotypes [95]. Where surveys covered AI tools broadly, findings were attributed specifically to GenAI only when the tool or generative use was identifiable [89–91]. Reported shortcomings included inaccurate or inconsistent answers, generic guidance, and code requiring correction, illustrating the need for verification and, in some contexts, instructor support [74,82,84,88].

The evidence therefore documented a broad range of academic assistance, most often directed by learners themselves. Perceived usefulness and measured learning outcomes did not consistently align, and the presence of tailored assistance did not establish its educational effectiveness. Comparative findings should be interpreted alongside the methodological limitations reported in Section 4.3.2. Section 4.5 examines the related evidence on creative agency and equitable learning.

4.5. Findings on Creative Agency and Equitable Learning (RQ2)

Evidence concerning creative agency was reported in 60 of the 61 included studies, and evidence concerning equitable learning in 35; 34 contributed to both dimensions. Themes were not mutually exclusive and included assessed outcomes, reported practices, perceptions, and concerns. Their frequencies indicate how often themes were reported, rather than the magnitude or certainty of effects. Percentages in Table 5 and Figure 3 use all 61 studies as the denominator.

Favorable creative-agency findings concerned critical, reflective, or problem-solving thinking (41 studies), autonomy and control over creative work (39), and idea generation (33). Students described selecting, adapting, or rejecting GenAI suggestions while retaining decisions about their work [56,82,88,100,108]. Evidence also included higher reflective-thinking scores with a Socratic conversational agent [50] and higher self-reported creativity and critical-thinking scores in a ChatGPT programming group [87]. A design study documented higher originality ratings and more creative-thinking activity during AI-assisted phases, alongside substantial direct adoption of generated content [104]. These findings distinguish assessed performance from perceived benefits and observed practices.

Favorable equitable-learning findings concerned disability-related accessibility (11 studies), linguistic support (ten), and targeted support for identifiable groups or learners with lower prior achievement (nine). Examples included assistance with reading, writing, and organization [67,78,82], and translation or academic-language support [53,54,89,105,108]. These findings did not necessarily establish reduced learning disparities. In particular, a dyslexia-focused survey did not record participants' diagnoses [91], and differentiated resources prepared by prospective teachers were not evaluated among their intended school pupils [56]. Table 5 summarizes the themes.

Overreliance and reduced independent thinking was reported in 39 studies, encompassing concerns about dependence and accounts of passive acceptance or limited revision of generated content [62,96,100,102]. In a writing experiment, participants reported lower

perceived control and ownership during ChatGPT-assisted writing; approximately 71% of their assisted introductions consisted of directly copied AI text [106]. These findings did not uniformly demonstrate deterioration in independent thinking. Eight studies were coded under concerns about voice, authorship, or ownership; linked reports from one disability-support survey presented differing summaries of concern about personal voice that could not be reconciled [78,79]. Other themes concerned originality, independent judgment, and unequal conditions constraining agency [48,58,77,95,99].

Table 5. Themes of creative agency and equitable learning reported in the included studies (n = 61).

Dimension	Reported Theme	Studies, *n* (%)	References
Creative agency	(+) Autonomy and control over creative work	39 (63.9)	[56,82,88,100–105,107,108]
	(+) Critical, reflective, and problem-solving thinking	41 (67.2)	[50,72,95,97,98,100,101, 103–105,107,108]
	(+) Idea generation and divergent thinking	33 (54.1)	[58,89,92,99,102–105,108]
	(−) Overreliance and reduced independent thinking	39 (63.9)	[62,67,90,96–102,104–107]
	(−) Loss of voice, authorship, or ownership	8 (13.1)	[64,78,86]
	(−) Difficulty exercising independent judgment with AI guidance	3 (4.9)	[58,77,83]
	(−) Reduced originality or homogenized outputs	8 (13.1)	[52,88,95,99]
	(−) Unequal conditions constraining creative agency	1 (1.6)	[48]
Equitable learning	(+) Accessibility and disability-related support	11 (18.0)	[67,78,91]
	(+) Targeted support for identifiable groups or learners with lower prior achievement	9 (14.8)	[53,59,93]
	(+) Linguistic support and reduced language barriers	10 (16.4)	[53,54,89,105,108]
	(−) Unequal access, cost, and digital divide	12 (19.7)	[48,66,79,105]
	(−) Barriers reported by identifiable learner groups	12 (19.7)	[67,78,91]
	(−) Unequal benefit across learner groups	7 (11.5)	[58,63,93]

Note: (+) denotes favorable findings and (−) unfavorable findings, including reported practices, perceptions, and concerns rather than necessarily measured changes in agency or educational disparities. Themes are not mutually exclusive. Percentages use 61 distinct studies; linked reports were combined, and each study was counted once per theme. References are illustrative.

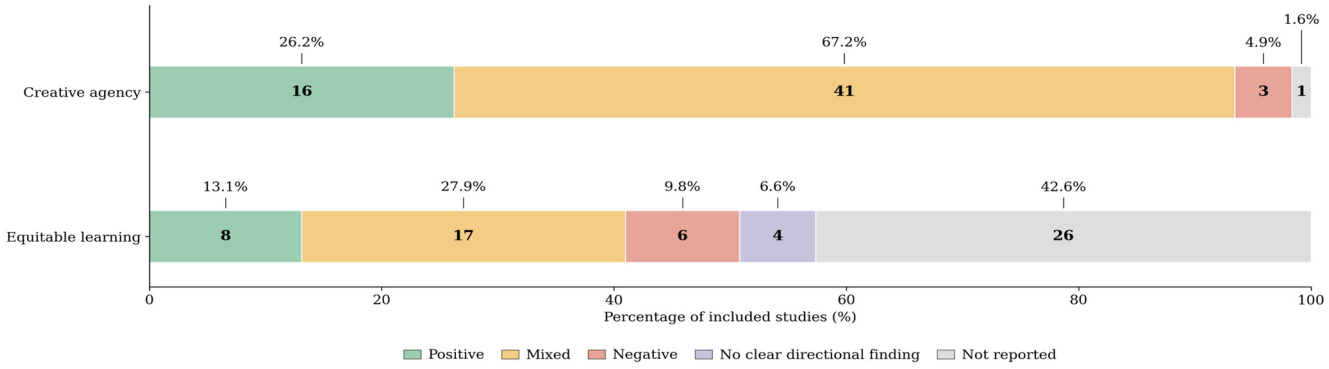


Figure 3. Direction of findings on creative agency and equitable learning across 61 studies. Segment labels indicate study counts; percentages use all 61 studies, including "Not reported." Directions were assigned after combining linked reports.

Unfavorable equitable-learning findings concerned barriers reported by identifiable learner groups (12 studies), unequal access, cost, or connectivity (12), and unequal benefits across groups (seven). Examples included interface difficulties among visually impaired

students [67], unclear institutional guidance [78,91], and payment barriers to more capable models [105]. Group differences concerned social class, disability, gender, and prior performance or proficiency [48,58,63,73,77,80]. Such differences did not consistently establish effects attributable to GenAI: assessment gaps varied across cohorts in electrotechnics [93], while another study found no significant interaction between baseline achievement and learning gains [97].

Mixed findings predominated for creative agency (41 studies), followed by positive (16) and negative findings (three); one study did not report this dimension. Equitable-learning findings were mixed in 17 studies, positive in eight, negative in six, and without a clear direction in four; 26 did not report this dimension (Figure 3).

Mixed classifications indicate coexisting favorable and unfavorable findings, without implying equal weight. These distributions do not quantify effect sizes or establish educational effectiveness. Section 4.6 examines connections to SDG 4 and SDG 10.

4.6. Connections to SDG 4 and SDG 10 (RQ3)

Connections to the Sustainable Development Goals were predominantly interpretive. Seven studies explicitly framed their aims, interpretation, or educational content in terms of SDG 4 [49,54,56,60,61,69,99], and one in terms of SDG 10 [49]. Only [49] named a specific target, 4.5; no study measured an official SDG indicator. Review-level connections were identified for 54 studies concerning SDG 4 and 32 concerning SDG 10. The remaining 28 studies did not support a connection to SDG 10. Figure 4 presents these mutually exclusive connection levels.

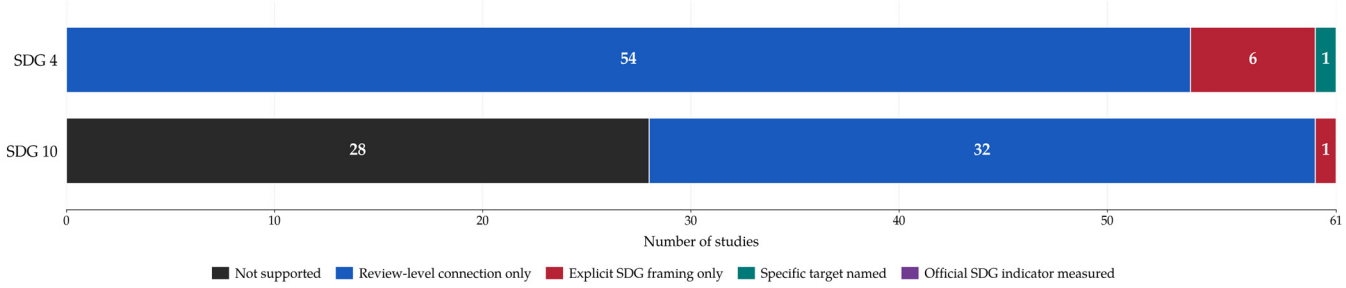


Figure 4. Highest documented connection to SDG 4 and SDG 10 across 61 studies. Each study appears once per goal. Categories distinguish unsupported connections, review-level interpretation, explicit author framing without a named target, and a specific target named. No study measured an official SDG indicator.

The clearest explicit target-level connection concerned target 4.5 [49]. Relevant findings included disability-related accessibility, linguistic assistance, and targeted support, alongside barriers and unequal benefits [48,63,67,78,89,91,93,105,108]. These findings concerned educational experiences and within-setting differences rather than demonstrated reductions in institutional or system-level disparities.

Connections to other targets were interpreted at the review level. Critical and problem-solving thinking related to target 4.4, while accessibility, language barriers, and unequal opportunities informed connections to targets 10.2 and 10.3. For target 4.7, one study reported increased sustainability knowledge following an intervention combining sustainability instruction, ChatGPT-assisted scenario development, and digital comics; its design could not isolate ChatGPT's contribution [99]. Table 6 summarizes these relationships and their limits.

Overall, the evidence concerned learning support, participation, and barriers within educational settings. Explicit SDG 10 framing was confined to one study [49]; subgroup comparisons did not establish inequality reduction attributable to GenAI. Sustainability knowledge was assessed in one combined intervention [99], but the findings did not

establish institutional or policy-level sustainability outcomes or progress toward official SDG targets. Section 4.7 examines variation across study characteristics.

Table 6. Interpretive relationship between reported findings and selected SDG targets.

SDG Target	Basis of Connection	Relevance of Reported Findings	Scope of the Available Evidence	References
4.4. Relevant skills	Review-level interpretation.	Critical, reflective, and problem-solving thinking; concerns about overreliance.	Educational skills, practices, and perceptions were examined; employment, decent-work, and entrepreneurship outcomes were not assessed.	[50,62,72,87, 90,94,97,101, 104,107]
4.5. Equal access and elimination of educational disparities	Review-level interpretation; target explicitly named in [49].	Disability-related accessibility, targeted and linguistic support; barriers and unequal benefits.	Within-setting differences were reported; assessment gaps and subgroup analyses did not establish reduced disparities attributable to GenAI.	[48,49,63,67, 78,89,91,93,97, 105,108]
4.7. Knowledge and skills for sustainable development	Review-level interpretation; explicit sustainability-education framing in [99].	Sustainability knowledge following a combined intervention; sustainability-related activities and green digital competences.	One study assessed sustainability knowledge without isolating ChatGPT's contribution. Other connections lacked dedicated sustainability-learning measures.	[60,64,99,103]
10.2. Social inclusion	Review-level interpretation	Accessibility and linguistic support; barriers experienced by identifiable learner groups.	Findings concerned educational participation and support; broader social or economic inclusion was not assessed.	[47,52,67,78, 82,89,91,105, 108]
10.3. Equal opportunity and reduced inequalities of outcome	Review-level interpretation	Cost, device, and connectivity barriers; targeted support and differences in educational benefit.	Group differences and descriptive assessment gaps did not establish causal reductions in institutional or population-level inequalities.	[48,58,63,73, 74,77,80,93,97, 105]

Note: Target descriptions paraphrase the United Nations 2030 Agenda. References illustrate the review-level mapping; studies may appear in multiple rows. Explicit sustainability-education framing does not imply that target 4.7 was named. These relationships do not indicate measured progress toward targets or official indicators.

4.7. Exploratory Subgroup Patterns

Exploratory study-level comparisons showed mixed creative-agency findings in 20 of 24 studies examining naturally occurring GenAI use and 21 of 37 examining implemented or structured activities. At least one unfavorable equitable-learning theme appeared in 13 and ten studies, respectively. All nine studies focusing on learners at risk of educational disadvantage reported an unfavorable equitable-learning theme; eight reported mixed creative-agency findings and one did not report creative agency [47,48,52,63,67,78,79,89,91,93]. Their focus on disadvantage may partly explain the greater visibility of barriers; this pattern does not establish more harmful GenAI effects in these populations.

Regional and disciplinary comparisons showed no uniform pattern. Mixed creative-agency findings appeared in four of ten studies using purpose-built or course-customized systems, compared with 37 of 51 other studies. Small, overlapping groups and differences in design and outcome coverage precluded attributing these variations to study characteristics. Section 4.8 examines the restricted sensitivity scenario.

4.8. Sensitivity Analysis

The sensitivity analysis temporarily omitted 39 of 61 studies (63.9%), retaining 22, including 16 retained with caution. Dimension-specific coverage decreased from 51 to 21 studies for learning individualization, 61 to 22 for student support, 60 to 22 for creative agency, 35 to 16 for equitable learning, 61 to 22 for SDG 4, and 33 to 15 for SDG 10. Among retained studies, creative-agency findings were mixed in 18 and positive in four. Equitable-

learning findings were mixed in eight, positive in four, negative in three, and without a clear direction in one.

Academic support and coexisting opportunities and concerns remained represented, but the restricted evidence was substantially narrower. Only one of five studies documenting system-driven adaptation and two of ten using purpose-built or course-customized systems remained. The sole study explicitly framing findings in terms of SDG 10 [49] was omitted, leaving only review-level connections; the combined intervention assessing sustainability knowledge [99] was also omitted. Persistence of themes indicates continued representation rather than confirmation of effectiveness or robustness. All 61 studies remained in the primary synthesis. Supplementary Material File S6 documents the decisions and dimension-level comparisons.

5. Discussion

This review highlights a gap between the educational support documented in GenAI use and the evidence needed to establish sustained learning, equity, and sustainability outcomes. Opportunities for assistance and creative participation coexisted with dependence and access barriers. Connections to SDG 4 and SDG 10 therefore concern educational processes and conditions, rather than demonstrated progress toward either goal.

5.1. Learning Individualization and Student Support (RQ1)

Learning individualization depended largely on students' choices: when to seek help, what to ask, and whether a response suited their task (Section 4.4; Table 4). Learner-directed individualization appeared in 46 of 61 studies, whereas system-driven adaptation was documented in five. These frequencies reflect the review's operational definitions and included evidence, rather than the wider literature. Support delivered on request, with use depending partly on task complexity and students' familiarity with the tool, was also described in a review of ChatGPT in higher education [109]. A review of university implementation cases identified differences in students' ability to formulate effective prompts [110]. Conversational responsiveness and course configuration do not themselves establish adaptation to each student's developing understanding. An educational support framework proposed using learning-management-system data and mastery rules for such adjustments, but its adaptive components underwent expert review rather than learner testing [111]. Integration into curricula, assessment, and teaching remains an identified research gap [112]. Only one of the five studies documenting system-driven adaptation remained in the restricted synthesis, limiting conclusions about its effectiveness.

The breadth of support contrasted with less consistent evidence of improved learning (Section 4.4; Figure 2). Perceived confidence, comprehension, or study strategies did not necessarily demonstrate independently assessed gains. Students' appreciation of immediate feedback and individual assistance coexisted with difficulties verifying unreliable responses in a scoping review [113]. A meta-analysis reported positive average effects on academic performance, while noting limited power analyses and concerns about post-intervention assessments [114]. A subsequent audit found that only four of 19 academic-performance comparisons adequately described both conditions and used an outcome measure representing learning [115]. Evaluations should therefore specify GenAI use and comparison conditions, distinguish assisted work from independent learning, and assess outcomes after assistance is withdrawn. Similarly, planning, reassurance, and perceived confidence should not be equated with systems that detect and respond to learners' affective states. Helping students verify responses and recognize when human guidance is needed remains an instructional priority.

5.2. Creative Agency and Equitable Learning (RQ2)

The central tension in creative agency was that opportunities to develop ideas and exercise control over work coexisted with concerns about reliance on generated answers and loss of independent judgment (Section 4.5; Table 5). Findings were mixed in 41 of the 60 studies reporting this dimension (Figure 3), without implying equal strength of favorable and unfavorable findings. A systematic review associated gains in critical and creative thinking with inquiry-oriented, scaffolded activities, while some studies of unstructured use reported weaker critical engagement or cognitive offloading [116]. Benefits from structured prompts and reflection, alongside risks of uncritical acceptance during unguided use, also emerged in a review focused on critical thinking [117]. The exploratory comparisons in this review found mixed findings more frequently in naturally occurring use than in implemented activities (Section 4.7), but these categories did not measure instructional guidance or establish its effects. Assessed originality, perceived creativity, observed revision practices, and concerns about dependence represent different forms of evidence (Section 4.5). In art education, unclear criteria for distinguishing human and machine contributions complicated assessments of individual creativity and originality [118]. Activities and assessments could therefore require students to evaluate generated suggestions, justify revisions, and make their contribution visible.

Equitable learning appeared to depend on whether students could translate GenAI support into meaningful benefit, rather than availability alone. Accessibility, targeted assistance, and linguistic support coexisted with access barriers and unequal benefits; findings were mixed in 17 of the 35 studies reporting equitable-learning evidence (Figure 3). The conceptual Divide Cascade distinguishes access, effective use, benefit realization, and durable learning, with disadvantage potentially accumulating across these stages [119]. An evidence map identified infrastructure, affordability, AI literacy, and gaps in equity-focused institutional policies as barriers to adoption [120]. All nine studies focusing on learners at risk of educational disadvantage reported an unfavorable equity theme (Section 4.7). Their focus may partly explain the visibility of barriers and does not establish greater harm from GenAI. Although several studies included students with disabilities, none focused specifically on motor disabilities; limited coverage of this group was also identified in an accessibility review [121]. A broader review argued that equitable outcomes require deliberate design and monitoring rather than adoption alone [122]. Institutions should therefore assess accessibility with affected learners, address costs and skills, and compare educational benefits across groups. Whether such measures reduce lasting learning disparities remains unestablished.

5.3. Implications for SDG 4 and SDG 10 (RQ3)

The clearest connection between the reviewed findings and SDG 4 concerned target 4.5: accessibility, linguistic assistance, and targeted support coexisted with barriers and unequal benefits (Section 4.6; Table 6). As contextual evidence outside the included synthesis, a faculty study linked accessibility and multilingual support to targets 4.5 and 4.a, emphasizing human review and institutionally funded access [123]. The included studies, however, did not establish reduced institutional disparities; contrasting assessment gaps among domestic and international students illustrate why support should not be equated with inequality reduction [93]. Connections to target 4.4 concerned critical thinking, creativity, and autonomous learning, consistent with competencies identified in another review [124], but employment-related outcomes were not assessed. For target 4.7, a combined intervention assessed sustainability knowledge without isolating ChatGPT's contribution [99]. A critical review similarly identified possibilities for sustainability education alongside access barriers, overreliance, unreliable outputs, and environmental costs [125]. The present

review addresses educational and social implications; it does not evaluate environmental or economic sustainability.

SDG 10 connections were particularly limited: one study provided explicit framing, 32 supported review-level interpretation, and 28 supported no connection; none measured an official indicator (Figure 4). The explicitly framed study [49] was omitted in sensitivity analysis, leaving only interpretive connections. Findings relevant to targets 10.2 and 10.3 concerned access, barriers, and differences in benefit within educational settings (Section 4.6; Table 6). A broader review described potential socioeconomic benefits alongside risks of uneven distribution and a widening educational digital divide [126]. Linking individual assistance to SDG 10 requires evidence of sustained, fairly distributed benefits from access arrangements, instructional support, and governance. A scoping review likewise called for comparative and longitudinal equity research [127]. The Divide Cascade offers a conceptual explanation of how access may fail to translate into durable benefits, but its propositions require empirical testing [119]. Future evaluations should specify SDG targets, compare learner groups over time, and examine institutional outcomes rather than infer them from individual experiences.

5.4. Limitations

The evidence was limited by frequent reliance on perceptions and self-reports, heterogeneous designs and measures, and scarce follow-up beyond a semester. Some reports examined AI tools broadly, complicating GenAI-specific attribution; a dyslexia-focused study did not collect participants' diagnoses [91]. The MMAT appraisal identified limitations in representativeness, confounding control, randomization reporting, and mixed-methods interpretation. Criterion 5.5 received 33 "No" and four "Can't tell" judgments across 37 mixed methods studies, so adherence of both components to their respective methodological standards was not established in any of them (Section 4.3.2). These component-level limitations constrain confidence in the resulting findings. Sensitivity analysis omitted 39 of 61 studies (63.9%); continued representation of themes among the 22 retained studies did not establish robustness, and methodological uncertainties remained (Section 4.8). Retention followed criterion-level rules rather than overall quality rankings. Reporting bias and overall certainty of evidence were not formally assessed, further limiting confidence in the synthesis.

The review process also has limitations. The search covered four sources, including two publisher collections, and was restricted to peer-reviewed English-language publications from 2022 to 2026. The two unretrieved reports introduce possible availability bias; their eligibility and potential contribution to equity-related findings could not be assessed. ScienceDirect and MDPI searches required creativity-related outcome terms and may have underrepresented studies focused on individualization, support, or equity without that vocabulary. No supplementary searches addressed this asymmetry. Additional studies may have appeared after the August 2026 search. Eligibility was narrower than in the registered protocol because studies had to provide evidence for at least one dimension of RQ1 and at least one dimension of RQ2, potentially excluding findings relevant to only one question or to the SDGs (Section 3.3). Theme and direction coding and SDG mapping involved interpretation; reported concerns did not necessarily indicate observed harm. RQ3 classifications were extracted by one reviewer and checked by a second, without independent dual coding or a calculated agreement statistic. Heterogeneous outcomes supported narrative synthesis rather than meta-analysis, while small and overlapping exploratory subgroups limited comparisons.

6. Conclusions

This systematic review of 61 studies represented by 63 reports distinguished GenAI-supported assistance in higher education from demonstrated learning and equity outcomes. Individualization largely reflected learner control, while system-driven adaptation was less frequently documented. Opportunities for explanation, feedback, idea development, and accessible support coexisted with concerns about overreliance and unequal access or benefit. Themes remained represented in the restricted synthesis, but substantial omissions and remaining methodological uncertainties precluded strong claims of robustness. Connections to SDG 4 and SDG 10 were predominantly interpretive and did not establish progress toward official targets. Although some studies assessed performance, the evidence did not establish sustained learning improvements or reductions in educational inequality attributable to GenAI. Educators should design activities that require students to verify and justify generated content, while institutions should address barriers to participation. Future research should distinguish assisted performance from independent learning and compare outcomes across learner groups over time. Studies linking GenAI to an SDG should specify the target and assess the educational and institutional outcomes relevant to it.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/su182211469/s1, Supplementary Material File S1: Reports excluded after full-text assessment and their principal reasons; Supplementary Material File S2: Included-studies data extraction matrix; Supplementary Material File S3: MMAT appraisal framework and operational guidance; Supplementary Material File S4: Independent MMAT ratings, comparisons, and consensus decisions; Supplementary Material File S5: Inter-rater agreement statistics; Supplementary Material File S6: Sensitivity-analysis decisions and study-level effects on the synthesis; Supplementary Material File S7: PRISMA 2020 checklist.

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Abbreviations

AC1	Gwet's first-order agreement coefficient
AI	Artificial intelligence
CI	Confidence interval
ERIC	Education Resources Information Center
EFL	English as a foreign language
GenAI	Generative artificial intelligence
INPLASY	International Platform of Registered Systematic Review and Meta-analysis Protocols
LLM	Large language model
MMAT	Mixed Methods Appraisal Tool
PICo	Population–Interest–Context
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PRISMA-P	Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols
RQ	Research question
SDG	Sustainable Development Goal
STEM	Science, technology, engineering, and mathematics
κ	Cohen's kappa coefficient

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